body like the earth revolving around a typical star like the sun, the warping of time actually has a far more significant impact on the earth's motion than does the warping of space.) We will return to a discussion of the warping of time after the next section.

Important as these three caveats are, so long as you hold them in the back of your mind, it is perfectly acceptable to invoke the warped-space image provided by the bowling ball on the rubber membrane as an intuitive summary of Einstein's new view of gravity.

Conflict Resolution

By introducing space and time as dynamic players, Einstein provided a clear conceptual image of how gravity works. The central question, though, is whether this reformulation of the gravitational force resolves the conflict with special relativity that afflicts Newton's theory of gravity. It does. Again, the rubber membrane analogy gives the essential idea. Imagine that we have a ball bearing rolling in a straight line along the flat membrane in the absence of the bowling ball. As we place the bowling ball on the membrane the motion of the ball bearing will be affected, but not instantaneously. If we were to film this sequence of events and view it in slow motion we would see that the disturbance caused by the introduction of the bowling ball spreads like ripples in a pond and eventually reaches the position of the ball bearing. After a short time, transitory oscillations along the rubber surface would settle down, leaving us with a static warped membrane.

The same is true for the fabric of space. When no mass is present, space is flat, and a small object will blissfully be at rest or will travel at a constant velocity. If a large mass comes on the scene, space will warp—but as in the case of the membrane, the distortion will not be instantaneous. Rather, it will spread outward from the massive body, ultimately settling down into a warped shape that communicates the gravitational pull of the new body. In our analogy, disturbances to the rubber membrane travel along its extent at a speed dictated by its particular material composition. In the real setting of general relativity, Einstein was able to calculate how fast disturbances to the fabric of the universe travel and he found that they travel at precisely the speed of light. This means, for instance, that in the hypothetical example discussed earlier in which the demise of the sun affects the earth by